

The Causal Effect of A Small Business Training Program on Business Outcomes

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Abstract

Causal inference shows larger revenue gains on average for small businesses (≤ 100 employees) who chose to participate in a business training program compared to those who did not. The back-doors that involved the observed variables were closed by controlling for the confounders by sub-classifying the firms into 12 classes based on 3 sectors and 4 revenue levels. To discount the effect of any general economic trends - during the training and evaluation period - that similarly affect both adopters and non-adopters, a difference-in-differences approach (between adopters and non-adopters before and after the training program) was used to provide an estimate of causation. The data did not indicate confounding effect of other pertinent observed variables such as (volatility in) number of employees, per-capita wage, wage bill to revenue ratio, service to sales revenue ratio. The subclassification ensured that these variables were not controlled, in order to close any back-doors arising from their potential collider effect. The question remains open whether useful finer grained causation estimates can be obtained involving these and other pertinent variables, such as duration of adoption, and higher order gradients in business outcome gains.

1 Introduction

High-quality management is critical for firm performance, and training programs are commonly implemented to strengthen managerial capabilities. Such programs typically cover human resources, interpersonal skills, marketing, and finance/accounting. For a small business (defined here as those with ≤ 100 employees), training in management practice is expected to have a significant impact (Busso et al 2023).

In order to evaluate the effectiveness of these programs, and the causal relationship between management trainings and business outcomes, we examine pre and post training business outcomes, for voluntary adopters and non-adopters of a Small Business Management Training program. Data are taken from monthly business-level employment information and outcome data from January 2010 to December 2019, where the program was first implemented in January 2013. For such voluntary adoption scenarios, the tested Quasi-experimental approach for causal inference is difference-in-differences, testing for robustness using regression. This factors out the effect of general economic trends affecting both adopters and non-adopters during the period of the training program.

In this paper, we aim to show, using careful causal inference, that there are larger revenue gains on average for small businesses who chose to participate in a business training program when compared to those who did not. There are approximately 180 small businesses that chose to participate in the business training program at some time during the testing period from January 2013 to December 2019. These businesses come from three sectors: Fast Food, Hospitality, and Retail.

2 Methodology

2.1 Data Pre-Processing

The panel dataset used for this study contained many reporting errors that could cloud the empirical results. The first issue arose with the firm IDs, where some firm IDs seemed to be identical except for an extra '00' at the end of their ID. Therefore these differences were consolidated and a total of 181 unique adopting firms were found from 2013 January to 2019 December. The second problematic

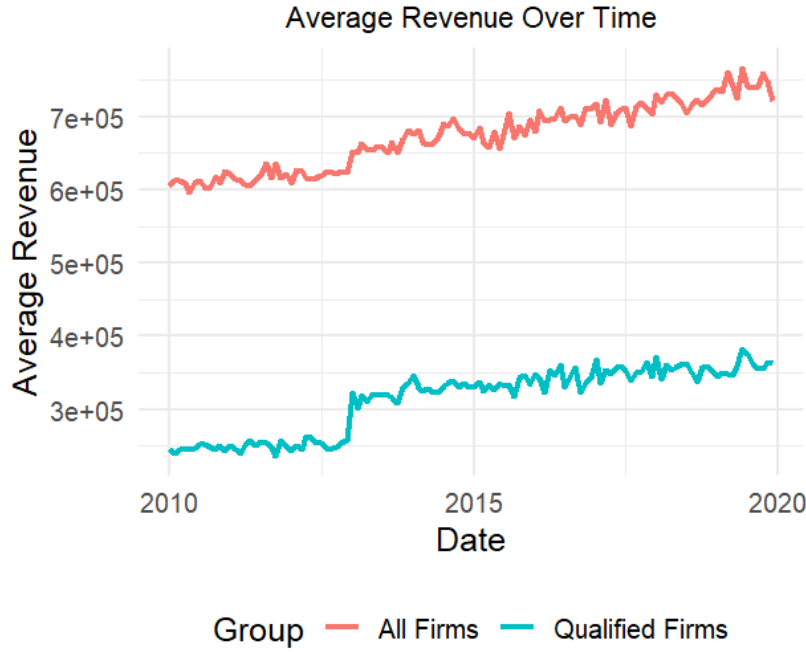


Figure 1: This Figure shows the difference in Average Revenue when we include unqualified firms, and when we remove them from our dataset.

reporting error was that many dates were not reported in clear date time format, which would cause problems when classifying groups as 'before and after' treatment. For these errors, it was determined that these dates could be replaced with the first day of the month (ie 2010-01-01) for each month.

The second data cleaning task was removing all rows of companies that had more than 100 employees (ie that were not qualified for the Small Business Management Training program). Keeping these unqualified firms in the non-adopters/control group would be a confounding factor as companies with larger number of employees are also likely to have a larger revenue, sales, or other variable that would remove credibility from a conclusion of the causal effect of the program.

2.2 Controlling for Confounding Variables by Subclassification

The real challenge in proving the causal relationship between adoption and improvement in business outcomes is to close back-doors, i.e. control the effect of several potential confounding variables that may influence both the choice to adopt and the business outcome (Cunningham 2021). Furthermore, controlling for these variables should not inadvertently open other back-doors, for example by controlling any potential collider variables. Such collider variables are affected both by adoption and improved business outcomes and controlling them could erroneously indicate causation where not even correlation exists.

The Directed Acyclic Graph in Figure 1 shows the underlying model of backdoor paths resulting from confounding observed variables (indicated by the data).

The Confounding Variables that were indicated by this dataset are:

1. The industry Sector (Fast Food, Retail, Hospitality),
2. The Revenue class (four classes, firms with revenue from 0-500k, 500k-1m, from 1m-1.5m, and from 1.5m-10m).

Back-doors involving confounding variables were closed by controlling these variables by *sub-classifying* firms into 12 classes based on 3 sectors and 4 revenue levels, while not opening any other back-doors involving potential collider variables.

We process the panel data of firm monthly revenues to assess the impact of adoption of the training program (at any time, for any nonzero length of time) on firm revenue. Firms with more than 100

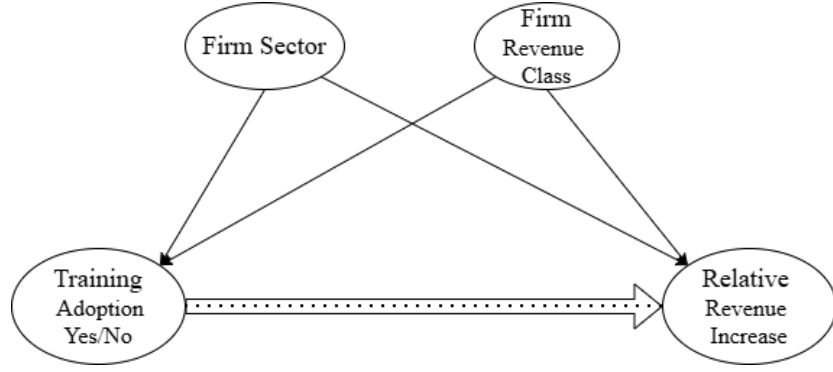


Figure 2: Directed Acyclic Graph for potential Causal relationship between Training Adoption and Relative Revenue Increase

employees in Jan 2013 at the start of training program are excluded. Each month's revenue data are split into the above mentioned 12 classes. Average revenue across firms in a class is tracked over time and adopter and non-adopter groups are compared.

2.3 The Difference in Differences Analysis

In order to prove causation between the intervention, i.e., adoption of the Small Business Training Program, and the outcome variable, we use a Difference-in-Differences (DD) model. From here, we focus on Average Revenue as the outcome variable. We represent our DD model as:

$$Y_{it} = \beta_0 + \beta_1 T_i + \beta_2 P_t + \beta_3 (T_i \times P_t) + \epsilon_{it}$$

Where:

- Y_{it} is the outcome (Revenue) for individual i in time period t .
- T_i is a dummy variable indicating the adopting group ($T_i = 1$ if adopt.t=1, $T_i = 0$ otherwise).
- P_t is a dummy variable indicating the post-treatment period ($P_t = 1$ if post-treatment, $P_t = 0$ if pre-treatment).
- β_0 is the intercept, representing the average outcome for the control group in the pre-treatment period. β_1 is the difference between the treatment and control groups in the pre-treatment period. β_2 is the change in the control group's outcome from the pre- to post-treatment period. β_3 is the DiD estimator, which measures the treatment effect.
- ϵ_{it} is the error term.

2.4 Robustness Check

The DD model assumes *Parallel Trends*, meaning that had the treatment not occurred, the difference between the two groups over time would have stayed the same. In Figure 3 we can visually find parallel trends between Control and Treatment groups, before treatment (from January 2010 to January 2013). However, we must also ensure that our treatment effect estimate is not sensitive to our DD model specifications, and that there are no unobserved variables in the treatment group, that change over time and affect the outcome.

In order to ensure robustness, we conduct a covariate sensitivity analysis by re-estimating the DD model while introducing new potential covariates. We begin by adding time-fixed effects to control any number of unobserved time-varying confounders.

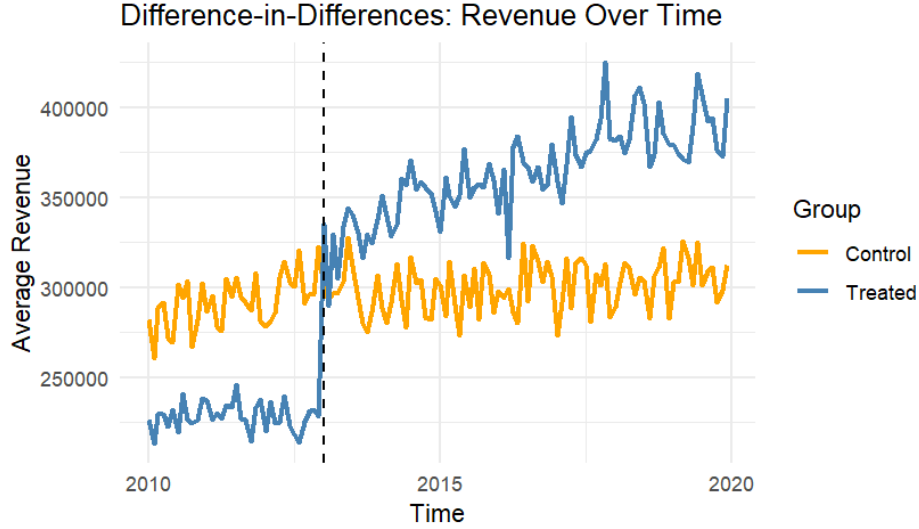


Figure 3: Diff in Diff representation combining all sectors and over all Revenue Classes; y axis shows average monthly revenue, while the x axis shows the time-points at the beginning of the training program and at the end of the evaluation period

3 Results

When we estimate the Difference-in-Differences model across all sectors and revenue classes, we obtain a positive coefficient on the interaction term, β_3 , indicating that adopting firms experienced larger increases in revenue relative to non-adopters following treatment. This estimate is statistically significant based on a one-tailed t-test.

We then estimate a Difference-in-Differences model of firm revenue including time fixed effects at the month level in order to control for common shocks affecting all firms. The coefficient on the interaction between treatment status and the post-adoption period is positive and statistically significant ($\beta = 127,090$; $p < 0.001$), indicating that adopting firms experienced an average increase in revenue of approximately 127,000 USD relative to non-adopting firms after adoption. The main effect of treatment status is negative, suggesting that treated firms had lower baseline revenues prior to adoption.

Therefore, as seen in Figure 3, this paper's 'primary' estimate of the program's causal affect is that over the evaluation period Average Monthly Revenue of adopters increased significantly while Average Monthly Revenue of non-adopters increased by less than 10,000 USD.

Figure 4 visualizes the potential causation of adoption on average firm revenue across the 12 class subclassification. This difference in differences shown before subclassification is sustained after subclassification for each sector and each revenue class. I.e., after controlling for these confounding factors, every subclass where comparison was possible showed the same trend of increasing revenue due to the program.

4 Conclusions

From Figures 3 and 4, we see an immediate jump in revenue immediately following the treatment period (January 2013) for the participating firms. This discontinuity raises the possibility that participating firms were already on different growth trajectories prior to program adoption. Thus, while the result of our Difference-in-Differences models are consistent with a statistically significant beneficial effect of the training program on Firm revenue, it should take into account the possibility of selection bias inherent in such a voluntary adoption scenario.

In addition, there are two potential methodological additions to this study that were considered but ultimately discarded. The first is integrating Two-Way fixed effects with differential timing. Because there are late-adopters of the program (ie after Jan 2013), the start date for each adopting firm could

have affected the outcome (revenue). However, out of the 181 adopting firms, 174 adopted in January 2013, meaning that only 7 adopted later, a negligible outlier group.

The second factor was the possibility of broadening the number of outcome variables that could be even more affected by the training program such as the ratio of sales to service revenue, ratio of wage bill to revenue, Wage Per Capita, or volatility in number of employees. In fact, although not confirmed by the data, these variables could be treated as potential collider variables in the model. The question remains open whether useful finer grained causation estimates can be obtained involving these and other pertinent variables, such as duration of adoption (instead of a Boolean variable), and higher order gradients in business outcome gains. These variables should be considered in further investigations, to recommend that the Small Business Training program should be advertised to more firms in the future.

References

- [1] Busso, M., Park, K., Irazoque, N. (2023). The Effectiveness of Management Training Programs: A Meta-Analytic Review. <https://doi.org/10.18235/0004815>
- [2] Cunningham, S. (2021). Causal inference: The mixtape. Yale University Press.

Average Revenue Over Time by Revenue Interval and Sector

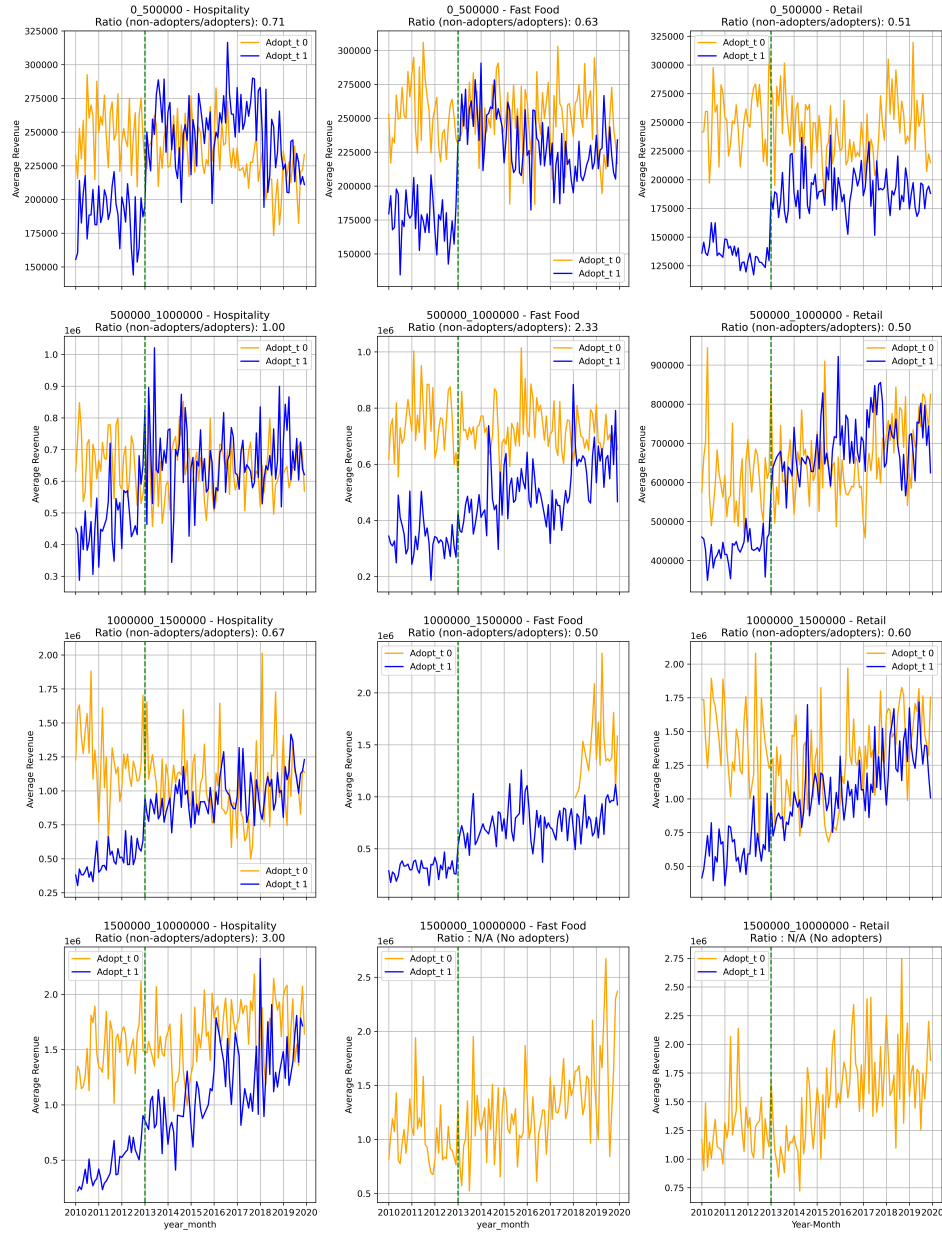


Figure 4: After the introduction of the training program (intervention, Jan 2013, green vertical line), adopting firms show higher revenue growth across sectors and revenue brackets. **Obs 0:** Firms with more than 100 employees in Jan 2013 are excluded since the training is meant for small firms. Firms with missing employee data are treated as eligible. **Obs 1:** Revenue intervals were based on data from 2019.11.csv, which had the highest number of firm IDs. In some cases (e.g., Fast Food in top revenue bracket), meaningful comparisons were not possible. **Obs 2:** Adopters generally had lower revenue at baseline (i.e., pre-2013). **Obs 3:** Plot titles include the ratio of non-adopters to adopters, determined using 2019 data. **Obs 4:** No non-adopters were found in the Fast Food sector in the \$1M–\$1.5M revenue class before 2017.